Task 11 – Final recommendation

Recommend a hash-table index keyed by unique student registration number for the stated workload, with a persistent database/index in a real system. The expected query cost is $\Theta(1)$, and the daily update schedule allows the index to be updated incrementally or rebuilt in a controlled batch. This is preferable to scanning up to one million records for every request. If ordered range queries, sorted reports or predictable worst-case guarantees are important, retain a sorted index/tree as an alternative. The recommendation assumes unique keys, adequate memory and a suitable hash implementation; the implementation must monitor collisions, load factor, persistence and consistency with the primary student database.